

Predicting Heart Disease with Machine Learning

Data cleaning, exploration and modelling on 270 patients

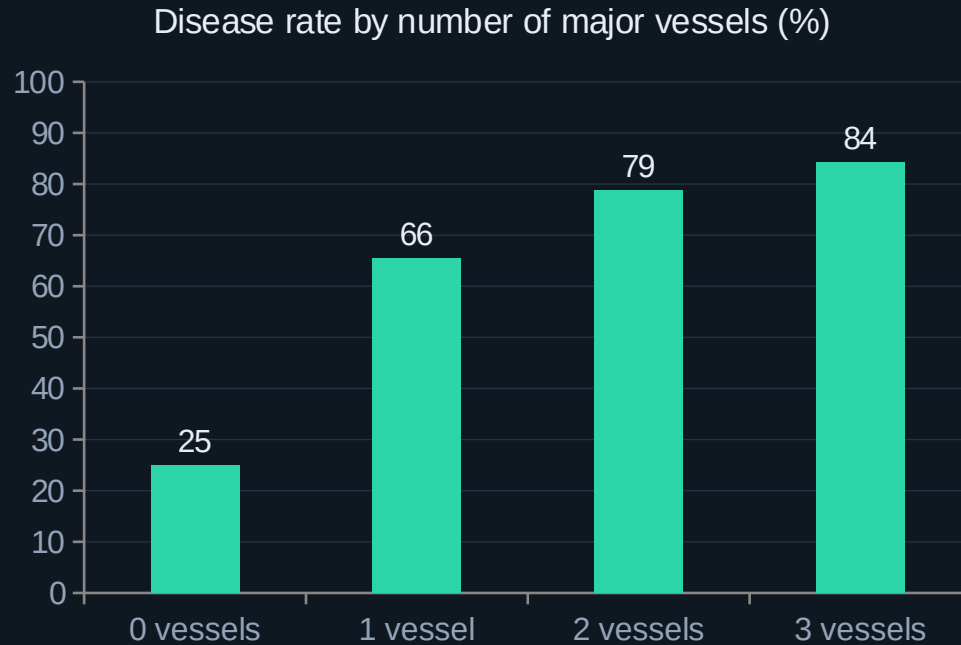
The problem and the data

- Goal: predict whether a patient has heart disease from routine medical measurements
- 270 patients, 13 features: age, sex, chest pain type, blood pressure, cholesterol, max heart rate, ECG and more
- Classes are fairly balanced: 150 healthy, 120 with disease
- Metric: accuracy, plus recall, because missing a sick patient is the costly mistake

Data cleaning

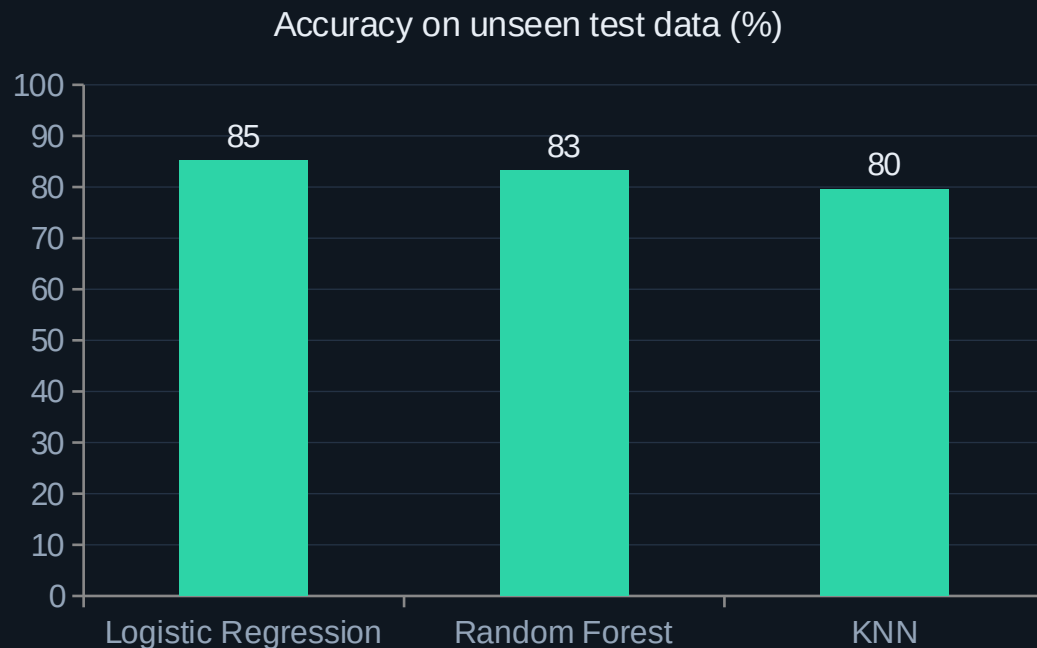
- Fixed column names (a stray space in "sex ") and renamed them clearly
- Recoded the target from 1/2 to 0/1
- Checked quality: 0 missing values and 0 duplicate rows
- Found a few outliers (9 in blood pressure, 5 in cholesterol) and kept them, because they are real clinical values, not errors

What relates to heart disease?



- Chest pain type 4: 70.5% have disease (vs about 17-25% for other types)
- Exercise-induced angina: 74% vs 30% without
- Higher max heart rate goes with less disease (correlation -0.42)

Model comparison



- Three models, same train/test split
- Logistic Regression was best and simplest
- Recall for patients with disease: 92%

Takeaways and limits

- A simple model reached 85% accuracy and caught 92% of patients who had the disease
- Most predictive signals: thal result, major vessels, exercise angina, ST depression and chest pain type
- Limits: only 270 patients and one test split, so results can vary
- This is a learning project, not a medical tool
- Next step: test on a larger dataset and tune the models